

**NATIONAL INSTITUTE OF TECHNOLOGY DURGAPUR**  
**Odd Semester End-term Examination, 2025-2026**

Course Code: CEC303

Full Marks: 60

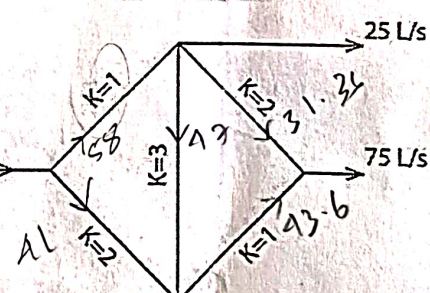
Course Name: Fluid Mechanic

Time: 3 hours

Question Paper No.: NITDGP/CEC303

Date of Exam: 19/11/2025

Instructions: Answer all the questions.

| Question No. | Body of the Question                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Marks | Mapped CO |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|
| 1            | <p>(a) List and explain the major steps involved in designing a water distribution network</p> <p>(b) What is the full form of CPHEEO, and as per the CPHEEO Manual, state:</p> <ul style="list-style-type: none"> <li>The minimum and maximum velocity to be adopted in the design of water distribution systems.</li> <li>The residual pressure at ferrule nodes for Class I, Class II, and other cities.</li> <li>The head loss gradient criteria and the minimum pipe diameter recommended.</li> </ul> <p>(c) What are the data requirements at nodes and pipes for running the EPANET model?</p> | 8     | CO3       |
| 2            | <p>The pressure difference <math>\Delta P</math> in a pipe of Diameter <math>D</math> and length <math>l</math> due to turbulent flow depends on the velocity <math>V</math>, viscosity <math>\mu</math>, density <math>\rho</math> and roughness <math>k</math>. Using Buckingham's <math>\pi</math> theorem, obtain an expression for <math>\Delta P</math>.</p>                                                                                                                                                                                                                                    | 8     | CO3       |
| 3            | <p>A 1:64 model is constructed of an open channel in concrete which has Manning's <math>n = 0.014</math>. Find the value of <math>n</math> for the model.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                         | 6     | CO1       |
| 4            | <p>Determine the distribution of flow rate in the figure shown below using the Hardy Cross method. Show the calculations up to second iteration.</p>                                                                                                                                                                                                                                                                                                                                                               | 8     | CO2       |
| 5            | <p>Define best hydraulic section and prove the condition under which the trapezoidal section of an open channel will be most economical.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1+4   | CO2       |
| 6            | <p>(a) Prove that in the case of a force vortex, the rise of the liquid level at the ends is equal to the fall of the liquid level at the axis of rotation.</p> <p>(b) Find the maximum speed of an open circular cylinder, having 180 mm diameter, 1200 mm length and containing water up to a height of 960 mm, at which it should be rotated about its vertical axis so that no water spill</p>                                                                                                                                                                                                    | 5 + 5 | CO1       |

**Course Outcomes**

CO1: Development of skills for predicting fluid behaviour

CO2: Knowledge of basics of fluid flow measurement and model development

CO3: Developing the requisite skill that helps in studying the advanced courses